

Latent Variable Moderated Mediation Using Correct and Misspecified Models

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Abstract

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Latent Variable Moderated Mediation Using Correct and Misspecified Models

Mediation analysis is a well-known procedure in the behavioral and social sciences (Charlton et al., 2021; Hayes, 2018; Pieters, 2017; Preacher et al., 2007). In its basic and standard form, the effect of an independent variable on an outcome is decomposed into a direct effect and an indirect effect transmitted through a mediator, with all variables observed. This basic single mediator model has since been extended in a number of directions, most relevant here being the case in which the indirect effect is allowed to depend on a moderator (Edwards & Lambert, 2007; Hayes, 2015; MacKinnon et al., 2007; Montoya, 2024; Preacher et al., 2007).

In this context, a fundamental concern arises from measurement. Many constructs in psychology are not directly observed but are measured indirectly through multiple items, that is, they are latent variables (Bollen & Hoyle, 2012). When such constructs are reduced to summary scores (e.g., a sum or mean of the items) and analyzed as though they were observed, measurement error is ignored. This error can bias the indirect effect and distort inferences about it (Cole & Preacher, 2014; Hoyle & Kenny, 1999), and it is especially problematic for the mediator, which acts at once as a predictor and an outcome (Gonzalez & MacKinnon, 2021). To address this, structural equation modelling (SEM) with latent variables has been proposed, especially in the context of mediation (Cheung & Lau, 2008; Falk & Biesanz, 2015; Ledgerwood & Shrout, 2011; Montoya, 2024). It estimates structural relationships between common factors, from which indicator-specific unique variance has been partitioned, provided the measurement model is correctly specified (Bollen, 1989; Rhemtulla et al., 2020). The problem is compounded under moderation, where the product term is even less reliable than either of its constituents (Bohrnstedt & Marwell, 1978; Busemeyer & Jones, 1983); yet we also note that latent variables do not necessarily improve power in general, whether for interactions or not (Cham et al., 2012; Fuller, 1987; Ledgerwood & Shrout, 2011; Wang & Rhemtulla, 2021).

Moderation mediation with latent variables is not straightforward in terms of statistical estimation. Indeed, a latent interaction involves the product of two latent variables, which is unobserved and nonnormal even when its constituents are normal (Kelava & Brandt, 2022).

Following this, a number of methods have been proposed to estimate latent interactions (e.g., Algina & Moulder, 2001; Arminger & Muthén, 1998; Asparouhov & Muthén, 2021; Boker et al., 2023; Bollen, 1995; Jöreskog & Yan, 1996; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Marsh et al., 2004; Mooijart & Bentler, 2010; Mooijart & Satorra, 2012; Wall & Amemiya, 2000). Most adopt a joint (“system-wide”) estimation strategy, in which the measurement and structural parameters are estimated simultaneously. This is the case of the known methods: latent moderated structural equations (LMS) and product-indicator (PI). Opposed to these are structural-after-measurement (SAM) approaches (Cox & Kelcey, 2021; Holst & Budtz-Jørgensen, 2020; Ng & Chan, 2020; Rosseel & Loh, 2024; Wall & Amemiya, 2001, 2003), in which the measurement model is estimated first and the structural model afterward with the measurement parameters held fixed. Many of the abovementioned approaches have so far been rarely applied in practice, likely held back by limited software, slow estimation, and error-prone implementations. With recent open-source implementations (Holst & Budtz-Jørgensen, 2020; Rosseel et al., 2025; Slupphaug et al., 2024) and non-commercial software (Keller, 2025), this is likely to change.

Beyond estimation, a further consideration concerns interpretation. Mediation is a hypothesis about causal effects, and causal interpretation requires identification assumptions that *may* not be verified from the data alone (Imai et al., 2010; Morelli et al., 2026; Rohrer et al., 2022; VanderWeele, 2015). With latent variable models, further caution is needed (Sterner et al., 2024; VanderWeele & Vansteelandt, 2022). We focus on one that is central to interpretation: the estimated latent variables must validly represent the intended constructs (Gonzalez & MacKinnon, 2021). In fact, when they do not, the model is subject to interpretational confounding (Burt, 1976), a discrepancy between the empirical meaning assigned to a latent variable through its estimated measurement parameters and the meaning intended by the researcher. Under system-wide estimation, the measurement and structural parameters are fit jointly, so the estimated loadings, and with them the empirical meaning of a latent variable, can change with the structural model that is imposed. The latent variable is then no longer anchored to its indicators

alone, and inferences about it need not be consistent across structural models. SAM avoids this by estimating the measurement model on its own. Thus, because the structural model cannot then influence it, the estimated latent variables, and their meaning, do not depend on the structure subsequently fit, and interpretational confounding is avoided. Note, however, that this does not imply the structure is correct (VanderWeele & Vansteelandt, 2022).

To illustrate interpretational confounding, we take a mediation model with three latent variables, X , M , and Y , each with three indicators, and obtain a population covariance matrix from a correctly specified version. We then alter it so that the covariances between the X and M indicators are unequal across the X indicators (3.0, 1.5, and 1.5, with mean held at the implied value of 2.0), changing only their pattern, not the overall X - M strength. To the same data we fit two structural models, one omitting the mediator and one including it, by system-wide SEM and by SAM ($N = 200$), and compare the estimated loadings of X . As shown in Table 1, the loadings of X estimated by system-wide SEM depend on the structural model: once the mediator enters, the loadings of x_2 and x_3 fall below their true value of one, even though the measurement of X is unchanged, so the empirical meaning of X shifts with the structure imposed. SAM estimates the measurement model from the indicators alone, so the X loadings are identical with and without the mediator and stay at their true values.

The example involves a single, deliberate change to the structural model, but it reflects a more general property of system-wide estimation. That is, because the measurement and structural parameters are estimated jointly, the consequences of a misspecification are not confined to the part of the model in which it occurs (Bainter & Bollen, 2014). Prior simulation work has examined latent interaction estimation mainly under distributional misspecification (Brandt et al., 2014; e.g., Cham et al., 2012; Lonati et al., 2024; Vieira et al., 2026), and in only one case under misspecifications at the measurement level (Brandt et al., 2020). Moreover, when it comes to (moderated) mediation, despite its popularity in applied research, only a few studies investigated it with latent variables, and those always assumed a correctly specified (and saturated) model (Falk & Biesanz, 2015; Feng et al., 2020).

We therefore consider three approaches in this article, two system-wide and one two-step. The first, LMS, approximates the nonnormal distribution of the latent product by a finite mixture of normals and estimates the joint model by full-information maximum likelihood (ML) via an expectation–maximization (EM) algorithm (Kelava et al., 2011; Klein & Moosbrugger, 2000). The second, the unconstrained PI approach (UPI), forms products of the observed indicators to serve as indicators of the latent interaction and estimates the resulting model by ML, optionally with robust standard errors and test statistics (Brandt et al., 2020; Kelava & Brandt, 2009; Lonati et al., 2024; Marsh et al., 2004; Slupphaug et al., 2024). The third, local structural-after-measurement (LSAM), proceeds in two steps: it first estimates the measurement model with a standard estimator (e.g., ML or generalized least squares) and then estimates the structural model with the measurement parameters held fixed. The second step is a path or regression model that can likewise be fit by various estimators (Rosseel et al., 2025; Rosseel & Loh, 2024). These approaches have been explained more in-depth in previous research (see Brandt et al., 2020; Kelava et al., 2011; Lin et al., 2010; Lonati et al., 2024; Rosseel et al., 2025; Slupphaug et al., 2024; Vieira et al., 2026).

Present Research and Contributions

This study evaluates LSAM, alongside LMS and UPI, for latent variable moderated mediation. We assess the three estimation approaches across a wide range of conditions and on a broad set of performance measures, both detailed in the next section.

Two questions motivate this evaluation. The first concerns the robustness of these approaches to structural and measurement misspecification, alone and in combination with nonnormal distributions, which is relevant because they rely on different distributional assumptions (e.g., Kelava et al., 2011; Lonati et al., 2024; Vieira et al., 2026). Connected to this, the second concerns how misspecification in the measurement model may affect the estimates, a question that is particularly relevant in moderated mediation, where the mediator functions simultaneously as an outcome and a predictor, and the effect of interest is a nonlinear function of two structural coefficients.

This evaluation makes three contributions to the literature. First, we provide a large-scale assessment of these approaches under misspecification of the measurement and structural models, which has not been done before. Second, we assess them for a derived quantity, the index of moderated mediation [imm; Hayes (2015)], a nonlinear function of the structural parameters whose behavior need not match that of its constituents and which has not previously been examined for LSAM. Third, whereas inference on such quantities has relied on the bootstrap (Hayes, 2015; Tibbe & Montoya, 2022), especially in the limited prior work concerning (moderated) mediation with latent variables (Falk & Biesanz, 2015; Feng et al., 2020), we adopt and evaluate the Monte Carlo method (MacKinnon et al., 2004; Preacher & Selig, 2012; Thoemmes et al., 2010), which may be accurate enough without the computational cost of the bootstrap, which can itself underperform at small sample sizes.

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate performance, following recent recommendations for the conduct and reporting of simulation studies (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024).

Population Model and Estimands

We study a moderated mediation model among four latent variables: an exogenous predictor X , a moderator W , a mediator M , and an outcome Y . The structural model is as follows

$$M = a_1X + a_2W + a_3(X \times W) + \zeta_M, \quad Y = bM + c'X + \zeta_Y,$$

The primary estimands are the interaction coefficient (a_3) and the imm (i.e., a_3b). The remaining structural coefficients (a_1, a_2, b, c') and the twelve freely estimated factor loadings serve as secondary estimands. Note the imm is a derived quantity rather than a free parameter, hence its true value is computed as a_3b implied by the correctly specified model.

Population values were fixed at $a_1 = 0.4$, $a_2 = 0.3$, $b = 0.5$, and $c' = 0.15$, with $\text{cor}(X, W) = 0.3$. The interaction coefficient a_3 is a design factor (see below). These coefficients, together with the structural disturbance variances, were chosen so that under the baseline

condition (normal exogenous predictors and a correctly specified model) each dependent latent variable has $R^2 = 0.30$ at the interaction level $a_3 = 0.20$. Note that, because a_3 contributes to the variance of M , the explained variance is naturally higher with a larger a_3 and lower when its effect is null. Each latent variable is measured by four congeneric indicators with loadings 1.0, 0.8, 0.7, and 0.6. Figure 1 shows the population model, where the structural misspecifications studied below are drawn in grey.

Simulation Conditions

We employed a fully factorial design crossing five factors: sample size ($N \in \{100, 200, 300, 500\}$), interaction magnitude ($a_3 \in \{0, 0.20, 0.40\}$), measurement reliability (low = 0.5, high = 0.7), the distribution of the exogenous latent predictors (normal, uniform, t , and two right-skewed χ^2 distributions), and the (mis)specification condition (nine levels; see below). This yielded $4 \times 3 \times 2 \times 5 \times 9 = 1,080$ conditions, each replicated 800 times.

Sample sizes entails the small-to-moderate range, where interaction terms are estimated with limited power (Brandt et al., 2020; Lonati et al., 2024). The interaction magnitudes include a null condition ($a_3 = 0$) to assess Type I error and false-positive coverage of the imm, and two nonzero levels representing small and moderate latent moderation effects. Reliability levels of 0.5 and 0.7 were chosen because measurement error is amplified in product terms and is therefore especially consequential for latent interaction estimation (Lonati et al., 2024; Vieira et al., 2026). Nonnormality of the latent predictors is both empirically common and methodologically central for latent interaction estimators, whose distributional assumptions differ across approaches (Cham et al., 2012; Klein & Moosbrugger, 2000; Klein & Muthén, 2007). The misspecification conditions follow the moderated-mediation framework for the two structural omissions (Preacher et al., 2007) and prior latent-interaction simulation work for the two measurement omissions (Brandt et al., 2020). They were also chosen as interpretable threats, acting respectively as a measured mediator–outcome confounder, a conflated moderated direct effect, exposure-correlated measurement error, and an unmeasured residual confounder (Edwards & Lambert, 2007; VanderWeele, 2015). The misspecification cases are detailed below.

Estimation Approaches

We compared a two-step estimator against two system-wide estimators. The two-step approach is LSAM (Rosseel et al., 2025; Rosseel & Loh, 2024), whose measurement model was estimated by ML in the first step. In this case, the interacting predictors X and W shared a single measurement block, while M and Y formed their own blocks. The two system-wide estimators are LMS (Klein & Moosbrugger, 2000), with robust standard errors, and the extended unconstrained product-indicator approach with double mean centering, which we refer to as UPI throughout, estimated with robust maximum likelihood (MLR), which combines Huber–White sandwich standard errors with a scaled (Yuan–Bentler) test statistic to accommodate the nonnormality of the indicator products (Kelava & Brandt, 2009; Lin et al., 2010; Marsh et al., 2004).

Data Generation

Appendix A describes the complete data-generating mechanism in detail. Briefly, the latent variables X and W were drawn from a bivariate distribution with $\text{Cor}(X, W) = \rho = 0.3$, constructed with the vine-to-anything (VITA) algorithm (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022). Five exogenous distributions were examined, each standardized to mean 0 and variance 1. Thus, the conditions differ only in distributional shape, not in scale or location. Specifically, the normal condition draws $X, W \sim \mathcal{N}(0, 1)$ through a forced Gaussian copula, whereas the remaining four combine non-normal margins with a non-Gaussian copula: uniform, t_5 , and two $\chi^2(2)$ conditions in which X and W are skewed in the same or in opposite directions. Given that, M and Y were generated from the structural equations above with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and $\zeta_Y \sim \mathcal{N}(0, \psi_Y)$, and the four indicators ($j = 1, \dots, 4$) of each latent variable $\eta \in \{X, W, M, Y\}$ were formed as $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$, with loadings $\lambda = (1, 0.8, 0.7, 0.6)$ and Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$ whose variances $\theta_{\eta j}$ were scaled to the target reliability $\omega \in \{0.5, 0.7\}$.

The nine (mis)specification conditions comprise one correctly specified condition and four families of misspecification, each at two empirically calibrated magnitudes (standardized 0.25 and 0.50, detailed in Appendix A). Two are structural: an omitted direct effect of the moderator on the

outcome (c2) and an omitted moderated direct effect (c3). The others are in the measurement model: a cross-loading of a mediator indicator on X (c1) and a residual covariance between a mediator and an outcome indicator (rc). In every case the effect is present in the population but omitted from the estimated model.

Nonconvergence, Outliers, and Inadmissible Solutions

Following previous research (Lonati et al., 2024; Vieira et al., 2026), replications were screened before analysis in two sequential steps, each applied globally so that every estimator is compared on an identical set of replications. First, we checked convergence: a fit was treated as nonconverged when any estimate, standard error, or confidence-interval bound—for any structural coefficient or loading—was missing or non-finite, and a replication was removed from all estimators whenever any one of them did not converge. We also flagged inadmissible solutions, where any residual (unique) variance was nonpositive or the latent covariance matrix was not positive definite. Second, among the remaining replications, we screened for outliers, identified per condition, method, and parameter as point estimates beyond the interquartile range ± 3 , together with replications whose estimated standard error exceeded 10 on any parameter of interest¹. Again, a replication flagged for any estimator was removed from all of them.

Performance Measures

Performance was assessed with the measures implemented in the *simhelpers* package (Joshi & Pustejovsky, 2025), computed for the *imm*, the structural coefficients, and the loadings. These measures, with their empirical and Monte Carlo standard error (MCSE) calculations, are listed in Table 2. Note the MCSE are obtained in closed form where available and by jackknife otherwise. On top of that, where applicable, we also report the rejection rate (Type I error when $a_3 = 0$, power otherwise).

Inference for the *imm* followed the Monte Carlo method as mentioned before: for $r = 1, \dots, R$ (with $R = 20,000$) we drew

¹ We added the standard error criterion because, even after excluding point estimate outliers, a single LMS replication, in one condition, retained an anomalous standard error; it was thus the only replication removed based on this criteria.

$$(a_3^{(r)}, b^{(r)}) \sim \mathcal{N}_2((\hat{a}_3, \hat{b}), \hat{\Sigma}_{a_3b}), \quad \text{imm}^{(r)} = a_3^{(r)} b^{(r)},$$

where $\hat{\Sigma}_{a_3b}$ is the estimated covariance of $(\hat{a}_3, \hat{b})$, and took the 2.5th and 97.5th empirical percentiles of $\{\text{imm}^{(r)}\}_{r=1}^R$ as the interval. The structural coefficients and loadings use Wald-based intervals from the model-implied parameter covariance matrix.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2025) using the *lavaan* (Rosseel, 2012) and *modsem* (Slupphaug et al., 2024) packages for estimation, *covsim* (Grønneberg et al., 2022) and *rvinecopulib* (Nagler & Vatter, 2025) for data generation, and *simhelpers* (Joshi & Pustejovsky, 2025) for performance evaluation. Conditions were run in parallel with the *parallel* package. Reproducible and independent random-number streams across workers were obtained with the L'Ecuyer-CMRG generator. The computational environment was pinned with Nix (Dolstra et al., 2004) via the *rix* package (Rodrigues & Baumann, 2026) to ensure reproducibility of the computational environment, and the article was written with the *apaquarto* extension (Schneider, 2026). For the package versions, code and materials, see the project repository.

Results

We present the results in three sections. We first report the imm and the interaction coefficient. We then turn to the remaining structural coefficients (i.e., a_1 , a_2 , b , and c'), and we close with the estimated factor loadings, focusing mainly the cross-loaded indicator. Moreover, due to the large number of conditions, we focused on a subset of the conditions. Namely, we fix the interaction coefficient at 0.2 (except for Type I error, which by construction requires a zero interaction) and display a limited number of sample sizes: $N = 200$ and 500 for the imm and the interaction coefficient, and solely $N = 500$ for the structural coefficients and loadings. For the latter two, we further restrict to high reliability, whereas the imm and the interaction coefficient are shown at both reliability levels. The entire set of results, across all sample sizes, reliabilities, and interaction values, is available in the repository (<https://doi.org/10.5281/zenodo.20703258>).

Main Structural Estimates

Relative Bias

Under correct specification with normal predictors, all three estimators recover both the γ and a_3 with negligible bias (Figure 2). In fact, bias is negligible across the nonnormal predictors as well, with the exception of the χ^2 conditions, where LMS and LSAM are the most affected. In this case, however, larger sample size removes the bias for LSAM but not for LMS, while higher reliability lessens it for both. Under misspecification, both c_2 and c_3 bias the γ for all three estimators, as expected. For a_3 , c_2 produces no bias, whereas c_3 leads to bias under the system-wide estimators. Among the measurement omissions, c_1 attenuates a_3 only under the system-wide estimators, while for the γ LMS shows bias in the case of χ^2 conditions. Last, the r_c leaves a_3 and the γ essentially unchanged. In all cases the bias increases with the magnitude of the omission and at lower reliability.

Note the bias of a_3 under c_3 is the one result not implied by the omission alone. As both omissions sit in the outcome equation, the mediator equation that defines a_3 stays correct. Hence, LSAM, which estimates a_3 from the mediator block alone, recovers it. The system-wide estimators fit the full covariance structure, where the latent interaction reaches the outcome only through a_3 and b , and absorb the omitted covariance into a_3 .

Relative RMSE

Under correct specification, relative RMSE falls with sample size and reliability for all three throughout distributions (Figure 3). Specifically, LMS is the most efficient, LSAM intermediate, and UPI the least. However, under misspecification, the structural omissions (c_2 and c_3) inflate the RMSE of the γ for all three estimators. In fact, such increase, in some cases, affects the system-wide estimators the most. With respect to the measurement omissions (c_1 and r_c), we observed a negligible effect. For a_3 , only c_3 inflates slightly the RMSE mainly for UPI, while LSAM and LMS are unaffected. The inflation is larger at lower reliability.

Coverage

Under correct specification with normal predictors, coverage is around the nominal level with all three approaches (Figure 4 and Figure 5) for the imm and a_3 . It goes down slightly under heavy-tailed (t_5) and χ^2 predictors for LMS and UPI, respectively, but mainly with a smaller sample size and low reliability.

Under misspecification, c2 and c3 reduce the imm coverage, since its interval is then centered on a biased estimate, with the largest drop for the system-wide estimators under c3. For a_3 , the cross-loading (c1) is the damaging case, collapsing coverage under the system-wide estimators at low reliability while LSAM holds near nominal. The remaining omissions leave coverage close to nominal.

Standard errors

Under correct specification the SE/SD ratio is near one for all three approaches (Figure 6 and Figure 7) in the case of a high reliability. Indeed, an overestimation is observed in case of low reliability and smaller sample size for UPI. Moreover, in the χ^2 in opposite directions, LSAM and UPI struggled the most throughout all misspecifications, including with the correct specification. These results were highly similar for both imm and a_3 . We also observe that the measurement misspecifications affect in particular LMS in combination with specific nonnormal distributions. Similar conclusions are obtained with the relative bias of the variance estimator (Figure 8).

Under misspecification the SE/SD ratio remains near one for all three. The variance-bias measure stays near one at high reliability (Figure 8) but becomes unstable at low reliability, where a few extreme replications inflate it for every method and most severely for LMS. Because these low-reliability cells reach values that would swamp the shared scale, they are shown separately in Figure B1, with the largest (above ten) truncated.

Hypothesis Testing (Type I error and Power)

Under correct specification with normal predictors and $a_3 = 0$, the imm test holds its nominal level reasonably well (Figure 9). UPI is slightly conservative, LSAM is closest to nominal, and LMS is liberal, with the same ordering for a_3 . The inflation of LMS grows under

nonnormal predictors, exceeding the nominal level in several skewed and heavy-tailed cells, while LSAM and UPI remain near it.

Under misspecification, $c3$ inflates the false-positive rate of the system-wide estimators, LMS the most, while LSAM holds its level. The cross-loading ($c1$) likewise inflates LMS, mainly at low reliability. The remaining omissions leave the rate near nominal.

At $a_3 = 0.2$ the index is detected with power that rises with sample size and reliability (Figure 10). LMS has the highest power throughout and UPI the lowest, with LSAM close to LMS, mirroring their efficiency ordering. Power falls under opposite-skew χ^2 predictors, dropping well below the conventional 0.80 at the smaller sample size, and most for UPI and LSAM. We note however that interpreting power in the case of bias and inflated Type I error should be avoided (Morris et al., 2019).

Additional Structural Estimates

For the remaining structural coefficients (a_1 , a_2 , b , and c'), at high reliability and $N = 500$, we report their relative bias (Figure 11), coverage (Figure 12), and SE/SD ratio (Figure 13) across all misspecification conditions and predictor distributions.

The coefficients show where each omission is absorbed. The two structural omissions act on the outcome equation: $c2$ biases the mediator-outcome slope b and, more strongly, the direct effect c' , and $c3$ is absorbed mainly by b . Under LSAM the mediator-equation coefficients a_1 and a_2 stay essentially unbiased under both structural omissions, since LSAM estimates the mediator block in isolation; the system-wide estimators, by contrast, leak $c2$ into the mediator equation, biasing a_2 (about 11% at the medium level) and, to a lesser extent, a_1 , and likewise carry part of $c3$ into c' . The cross-loading ($c1$) is the most disruptive, biasing all four coefficients for every estimator, because the omitted X loading on m_3 distorts the mediator's relations to the other latent variables. The residual covariance (rc) slightly inflates b and biases c' downward for all three. Across conditions the bias is consistently smaller under LSAM than under the system-wide estimators.

Taken together, under correct specification the three estimators recover the imm and a_3

with little bias and near-nominal coverage, and differ mainly in a calibration and efficiency trade-off. LMS is the most efficient but runs liberal and is the most sensitive to nonnormality. LSAM is the best calibrated in both coverage and Type I error at a modest efficiency cost. UPI is intermediate but unstable at the smallest sample size.

Measurement Loadings

For the twelve freely estimated factor loadings, all but one are recovered with near-nominal performance across every measure and condition. Hence, we display only the cross-loaded mediator indicator m_3 , with its relative bias, coverage, SE/SD ratio, and relative bias of the variance estimator shown together at high reliability and $N = 500$ (Figure 14). For m_3 , the omitted cross-loading (c1) is the only consequential misspecification: its relative bias rises sharply (above 0.5 at the medium level, somewhat less for LSAM) and its coverage collapses toward zero. The two standard error measures, by contrast, remain well calibrated: across conditions the SE/SD ratio and the relative bias of the variance estimator both stay near one, deviating only mildly under the strongest cross-loading, and the most so for LSAM. We also highlight that a subtler pattern, negligible in size, appears under the residual covariance (rc): the system-wide estimators carry a small systematic upward bias in m_3 , well under one percent, whereas LSAM stays centered on zero, since it estimates the mediator's measurement block in isolation.

Discussion

We evaluated three estimators of a latent variable moderated mediation model, the two-step LSAM and the system-wide LMS and UPI, under correct specification and under a large number of structural and measurement misspecifications. Two questions organized this study [...]

Under correct specification [...]

The estimators diverge under misspecification [...]

For practical recommendations [...]

The study has limits that bound these conclusions [...]

Several extensions follow naturally [...]

References

- Algina, J., & Moulder, B. C. (2001). A note on estimating the Jöreskog–Yang model for latent variable interaction using LISREL 8.3. *Structural Equation Modeling: A Multidisciplinary Journal*, 8(1), 40–52. https://doi.org/10.1207/S15328007SEM0801_3
- Arminger, G., & Muthén, B. O. (1998). A Bayesian approach to nonlinear latent variable models using the Gibbs sampler and the Metropolis–Hastings algorithm. *Psychometrika*, 63(3), 271–300. <https://doi.org/10.1007/bf02294856>
- Asparouhov, T., & Muthén, B. (2021). Bayesian estimation of single and multilevel models with latent variable interactions. *Structural Equation Modeling: A Multidisciplinary Journal*, 28(2), 314–328. <https://doi.org/10.1080/10705511.2020.1761808>
- Bainter, S. A., & Bollen, K. A. (2014). Interpretational confounding or confounded interpretations of causal indicators? *Measurement: Interdisciplinary Research and Perspectives*, 12(4), 125–140. <https://doi.org/10.1080/15366367.2014.968503>
- Bohrnstedt, G. W., & Marwell, G. (1978). The reliability of products of two random variables. *Sociological Methodology*, 9, 254–273. <https://doi.org/10.2307/270812>
- Boker, S. M., Oertzen, T. von, Pritikin, J. N., Hunter, M. D., Brick, T. R., Brandmaier, A. M., & Neale, M. C. (2023). Products of variables in structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, 30(5), 708–718. <https://doi.org/10.1080/10705511.2022.2141749>
- Bollen, K. A. (1989). *Structural equations with latent variables*. John Wiley & Sons. <https://doi.org/10.1002/9781118619179>
- Bollen, K. A. (1995). Structural equation models that are nonlinear in latent variables: A least-squares estimator. *Sociological Methodology*, 25, 223–251. <https://doi.org/10.2307/271068>
- Bollen, K. A., & Hoyle, R. H. (2012). Latent variables in structural equation modeling. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (pp. 56–67). The Guilford Press.
- Brandt, H., Kelava, A., & Klein, A. G. (2014). A simulation study comparing recent approaches

for the estimation of nonlinear effects in SEM under the condition of nonnormality. *Structural Equation Modeling: A Multidisciplinary Journal*, 21(2), 181–195.

<https://doi.org/10.1080/10705511.2014.882660>

Brandt, H., Umbach, N., Kelava, A., & Bollen, K. A. (2020). Comparing estimators for latent interaction models under structural and distributional misspecifications. *Psychological Methods*, 25(3), 321–345. <https://doi.org/10.1037/met0000231>

Burt, R. S. (1976). Interpretational confounding of unobserved variables in structural equation models. *Sociological Methods & Research*, 5(1), 3–52.

<https://doi.org/10.1177/004912417600500101>

Busemeyer, J. R., & Jones, L. E. (1983). Analysis of multiplicative combination rules when the causal variables are measured with error. *Psychological Bulletin*, 93(3), 549–562.

<https://doi.org/10.1037/0033-2909.93.3.549>

Cham, H., West, S. G., Ma, Y., & Aiken, L. S. (2012). Estimating latent variable interactions with nonnormal observed data: A comparison of four approaches. *Multivariate Behavioral Research*, 47(6), 840–876. <https://doi.org/10.1080/00273171.2012.732901>

Charlton, A., Montoya, A. K., Price, J., Hilgard, J., & Krefeld-Schwalb, A. (2021). *Noise in the process: An assessment of the evidential value of mediation effects in marketing journals*.

PsyArXiv preprint. <https://doi.org/10.31234/osf.io/ck2r5>

Cheung, G. W., & Lau, R. S. (2008). Testing mediation and suppression effects of latent variables: Bootstrapping with structural equation models. *Organizational Research Methods*, 11(2), 296–325. <https://doi.org/10.1177/1094428107300343>

Cole, D. A., & Preacher, K. J. (2014). Manifest variable path analysis: Potentially serious and misleading consequences due to uncorrected measurement error. *Psychological Methods*, 19(2), 300–315. <https://doi.org/10.1037/a0033805>

Cox, K., & Kelcey, B. (2021). Croon's bias-corrected estimation of latent interactions. *Structural Equation Modeling: A Multidisciplinary Journal*, 28(6), 863–874.

<https://doi.org/10.1080/10705511.2021.1922283>

- Dolstra, E., De Jonge, M., & Visser, E. (2004). Nix: A safe and policy-free system for software deployment. *18th Large Installation System Administration Conference*, 79–92.
- Edwards, J. R., & Lambert, L. S. (2007). Methods for integrating moderation and mediation: A general analytical framework using moderated path analysis. *Psychological Methods*, 12(1), 1–22. <https://doi.org/10.1037/1082-989X.12.1.1>
- Falk, C. F., & Biesanz, J. C. (2015). Inference and interval estimation methods for indirect effects with latent variable models. *Structural Equation Modeling: A Multidisciplinary Journal*, 22(1), 24–38. <https://doi.org/10.1080/10705511.2014.935266>
- Feng, Q., Song, Q., Zhang, L., Zheng, S., & Pan, J. (2020). Integration of moderation and mediation in a latent variable framework: A comparison of estimation approaches for the second-stage moderated mediation model. *Frontiers in Psychology*, 11, 2167. <https://doi.org/10.3389/fpsyg.2020.02167>
- Foldnes, N., & Grønneberg, S. (2015). How general is the Vale–Maurelli simulation approach? *Psychometrika*, 80(4), 1066–1083. <https://doi.org/10.1007/s11336-014-9414-0>
- Fuller, W. A. (1987). *Measurement error models*. Wiley. <https://doi.org/10.1002/9780470316665>
- Gonzalez, O., & MacKinnon, D. P. (2021). The measurement of the mediator and its influence on statistical mediation conclusions. *Psychological Methods*, 26(1), 1–17. <https://doi.org/10.1037/met0000263>
- Grønneberg, S., Foldnes, N., & Marcoulides, K. M. (2022). covsim: An R package for simulating non-normal data for structural equation models using copulas. *Journal of Statistical Software*, 102(3), 1–45. <https://doi.org/10.18637/jss.v102.i03>
- Hayes, A. F. (2015). An index and test of linear moderated mediation. *Multivariate Behavioral Research*, 50(1), 1–22. <https://doi.org/10.1080/00273171.2014.962683>
- Hayes, A. F. (2018). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach* (2nd ed.). The Guilford Press.
- Holst, K. K., & Budtz-Jørgensen, E. (2020). A two-stage estimation procedure for non-linear structural equation models. *Biostatistics*, 21(4), 676–691.

<https://doi.org/10.1093/biostatistics/kxy082>

Hoyle, R. H., & Kenny, D. A. (1999). Sample size, reliability, and tests of statistical mediation. In

R. H. Hoyle (Ed.), *Statistical strategies for small sample research* (pp. 195–222). Sage.

Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis.

Psychological Methods, 15(4), 309–334. <https://doi.org/10.1037/a0020761>

Jöreskog, K. G., & Yan, F. (1996). Nonlinear structural equation models: The Kenny–Judd model

with interaction effects. In G. A. Marcoulides & R. E. Schumacker (Eds.), *Advanced structural equation modeling* (pp. 57–88). Lawrence Erlbaum Associates.

Joshi, M., & Pustejovsky, J. (2025). *simhelpers: Helper functions for simulation studies*.

<https://CRAN.R-project.org/package=simhelpers>

Kelava, A., & Brandt, H. (2009). Estimation of nonlinear latent structural equation models using the extended unconstrained approach. *Review of Psychology*, 16(2), 123–131.

Kelava, A., & Brandt, H. (2022). Latent interaction effects. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (pp. 427–446). Guilford Press.

Kelava, A., Werner, C. S., Schermelleh-Engel, K., Moosbrugger, H., Zapf, D., Ma, Y., Cham, H., Aiken, L. S., & West, S. G. (2011). Advanced nonlinear latent variable modeling: Distribution analytic LMS and QML estimators of interaction and quadratic effects. *Structural Equation Modeling: A Multidisciplinary Journal*, 18(3), 465–491.

<https://doi.org/10.1080/10705511.2011.582408>

Keller, B. T. (2025). A factored regression approach to modeling latent variable interactions and nonlinear effects. *Psychological Methods*. <https://doi.org/10.1037/met0000777>

Kenny, D. A., & Judd, C. M. (1984). Estimating the nonlinear and interactive effects of latent variables. *Psychological Bulletin*, 96(1), 201–210.

<https://doi.org/10.1037/0033-2909.96.1.201>

Klein, A. G., & Moosbrugger, H. (2000). Maximum likelihood estimation of latent interaction effects with the LMS method. *Psychometrika*, 65(4), 457–474.

<https://doi.org/10.1007/BF02296338>

- Klein, A. G., & Muthén, B. O. (2007). Quasi-maximum likelihood estimation of structural equation models with multiple interaction and quadratic effects. *Multivariate Behavioral Research*, 42(4), 647–673.
- Ledgerwood, A., & Shrout, P. E. (2011). The trade-off between accuracy and precision in latent variable models of mediation processes. *Journal of Personality and Social Psychology*, 101(6), 1174–1188. <https://doi.org/10.1037/a0024776>
- Lin, G.-C., Wen, Z., Marsh, H. W., & Lin, H.-S. (2010). Structural equation models of latent interactions: Clarification of orthogonalizing and double-mean-centering strategies. *Structural Equation Modeling: A Multidisciplinary Journal*, 17(3), 374–391. <https://doi.org/10.1080/10705511.2010.488999>
- Lonati, S., Rönkkö, M., & Antonakis, J. (2024). Normality assumption in latent interaction models. *Psychological Methods*. <https://doi.org/10.1037/met0000657>
- MacKinnon, D. P., Fairchild, A. J., & Fritz, M. S. (2007). Mediation analysis. *Annual Review of Psychology*, 58, 593–614. <https://doi.org/10.1146/annurev.psych.58.110405.085542>
- MacKinnon, D. P., Lockwood, C. M., & Williams, J. (2004). Confidence limits for the indirect effect: Distribution of the product and resampling methods. *Multivariate Behavioral Research*, 39(1), 99–128. https://doi.org/10.1207/s15327906mbr3901_4
- Marsh, H. W., Wen, Z., & Hau, K.-T. (2004). Structural equation models of latent interactions: Evaluation of alternative estimation strategies and indicator construction. *Psychological Methods*, 9(3), 275–300. <https://doi.org/10.1037/1082-989X.9.3.275>
- Montoya, A. K. (2024). Combining statistical and causal mediation analysis. In H. T. Reis, T. West, & C. M. Judd (Eds.), *Handbook of research methods in social and personality psychology* (pp. 622–652). Cambridge University Press.
- Mooijjaart, A., & Bentler, P. M. (2010). An alternative approach for nonlinear latent variable models. *Structural Equation Modeling: A Multidisciplinary Journal*, 17(3), 357–373. <https://doi.org/10.1080/10705511.2010.488997>
- Mooijjaart, A., & Satorra, A. (2012). Moment testing for interaction terms in structural equation

- modeling. *Psychometrika*, 77(1), 65–84. <https://doi.org/10.1007/s11336-011-9232-6>
- Morelli, S., Faleh, R., & Brandt, H. (2026). *RAPSEM: Identifying latent mediators without sequential ignorability via a rank-preserving structural equation model*. arXiv preprint arXiv:2509.23935. <https://doi.org/10.48550/arXiv.2509.23935>
- Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11), 2074–2102. <https://doi.org/10.1002/sim.8086>
- Nagler, T., & Vatter, T. (2025). *rvinecopulib: High performance algorithms for vine copula modeling*. <https://CRAN.R-project.org/package=rvinecopulib>
- Ng, J. C. K., & Chan, W. (2020). Latent moderation analysis: A factor score approach. *Structural Equation Modeling: A Multidisciplinary Journal*, 27(4), 629–648. <https://doi.org/10.1080/10705511.2019.1664304>
- Pawel, S., Bartoš, F., Siepe, B. S., & Lohmann, A. (2025). Handling missingness, failures, and non-convergence in simulation studies: A review of current practices and recommendations. *The American Statistician*, 1–18. <https://doi.org/10.1080/00031305.2025.2540002>
- Pieters, R. (2017). Meaningful mediation analysis: Plausible causal inference and informative communication. *Journal of Consumer Research*, 44(3), 692–716. <https://doi.org/10.1093/jcr/ucx081>
- Preacher, K. J., Rucker, D. D., & Hayes, A. F. (2007). Addressing moderated mediation hypotheses: Theory, methods, and prescriptions. *Multivariate Behavioral Research*, 42(1), 185–227. <https://doi.org/10.1080/00273170701341316>
- Preacher, K. J., & Selig, J. P. (2012). Advantages of monte carlo confidence intervals for indirect effects. *Communication Methods and Measures*, 6(2), 77–98. <https://doi.org/10.1080/19312458.2012.679848>
- R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Rhemtulla, M., Bork, R. van, & Borsboom, D. (2020). Worse than measurement error:

- Consequences of inappropriate latent variable measurement models. *Psychological Methods*, 25(1), 30–45. <https://doi.org/10.1037/met0000220>
- Rodrigues, B., & Baumann, P. (2026). *Rix: Reproducible data science environments with Nix*. <https://CRAN.R-project.org/package=rix>
- Rohrer, J. M., Hünermund, P., Arslan, R. C., & Elson, M. (2022). That’s a lot to process! Pitfalls of popular path models. *Advances in Methods and Practices in Psychological Science*, 5(2), 1–14. <https://doi.org/10.1177/25152459221095827>
- Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Rosseel, Y., Burghgraeve, E., Loh, W. W., & Schermelleh-Engel, K. (2025). Structural after measurement (SAM) approaches for accommodating latent quadratic and interaction effects. *Behavior Research Methods*, 57(4), 101. <https://doi.org/10.3758/s13428-024-02532-y>
- Rosseel, Y., & Loh, W. W. (2024). A structural after measurement approach to structural equation modeling. *Psychological Methods*, 29(3), 561–588. <https://doi.org/10.1037/met0000503>
- Schneider, W. J. (2026). *apaquarto: A Quarto extension for creating APA style (7th edition) documents*. <https://github.com/wjschne/apaquarto>
- Siepe, B. S., Bartoš, F., Morris, T. P., Boulesteix, A.-L., Heck, D. W., & Pawel, S. (2024). Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting. *Psychological Methods*. <https://doi.org/10.1037/met0000695>
- Slupphaug, K. S., Mehmetoglu, M., & Mittner, M. (2024). modsem: An R package for estimating latent interactions and quadratic effects. *Structural Equation Modeling: A Multidisciplinary Journal*. <https://doi.org/10.1080/10705511.2024.2417409>
- Sterner, P., Pargent, F., Deffner, D., & Goretzko, D. (2024). A causal framework for the comparability of latent variables. *Structural Equation Modeling: A Multidisciplinary Journal*, 31(5), 747–758. <https://doi.org/10.1080/10705511.2024.2339396>
- Thoemmes, F., MacKinnon, D. P., & Reiser, M. R. (2010). Power analysis for complex

- mediational designs using monte carlo methods. *Structural Equation Modeling: A Multidisciplinary Journal*, 17(3), 510–534. <https://doi.org/10.1080/10705511.2010.489379>
- Tibbe, T. D., & Montoya, A. K. (2022). Correcting the bias correction for the bootstrap confidence interval in mediation analysis. *Frontiers in Psychology*, 13, 810258. <https://doi.org/10.3389/fpsyg.2022.810258>
- VanderWeele, T. J. (2015). *Explanation in causal inference: Methods for mediation and interaction*. Oxford University Press.
- VanderWeele, T. J., & Vansteelandt, S. (2022). A statistical test to reject the structural interpretation of a latent factor model. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 84(5), 2032–2054. <https://doi.org/10.1111/rssb.12555>
- Vieira, F. F., Slupphaug, K. S., & Rosseel, Y. (2026). *Evaluating local structural-after-measurement (LSAM) and traditional approaches for the estimation of complex nonlinear effects among latent variables*. PsyArXiv preprint. https://osf.io/preprints/psyarxiv/xk8um_v3
- Wall, M. M., & Amemiya, Y. (2000). Estimation for polynomial structural equation models. *Journal of the American Statistical Association*, 95(451), 929–940. <https://doi.org/10.2307/2669475>
- Wall, M. M., & Amemiya, Y. (2001). Generalized appended product indicator procedure for nonlinear structural equation analysis. *Journal of Educational and Behavioral Statistics*, 26(1), 1–29. <https://doi.org/10.3102/10769986026001001>
- Wall, M. M., & Amemiya, Y. (2003). A method of moments technique for fitting interaction effects in structural equation models. *British Journal of Mathematical and Statistical Psychology*, 56(1), 47–63. <https://doi.org/10.1348/000711003321645331>
- Wang, Y. A., & Rhemtulla, M. (2021). Power analysis for parameter estimation in structural equation modeling: A discussion and tutorial. *Advances in Methods and Practices in Psychological Science*, 4(1), 1–17. <https://doi.org/10.1177/2515245920918253>

Table 1*Illustration for Interpretational Confounding.*

	Parameter	True	SEM, no M	SEM, with M	SAM, no M	SAM, with M
x1	Loading of x_1	1	1	1.00	1	1
x2	Loading of x_2	1	1	0.85	1	1
x3	Loading of x_3	1	1	0.85	1	1

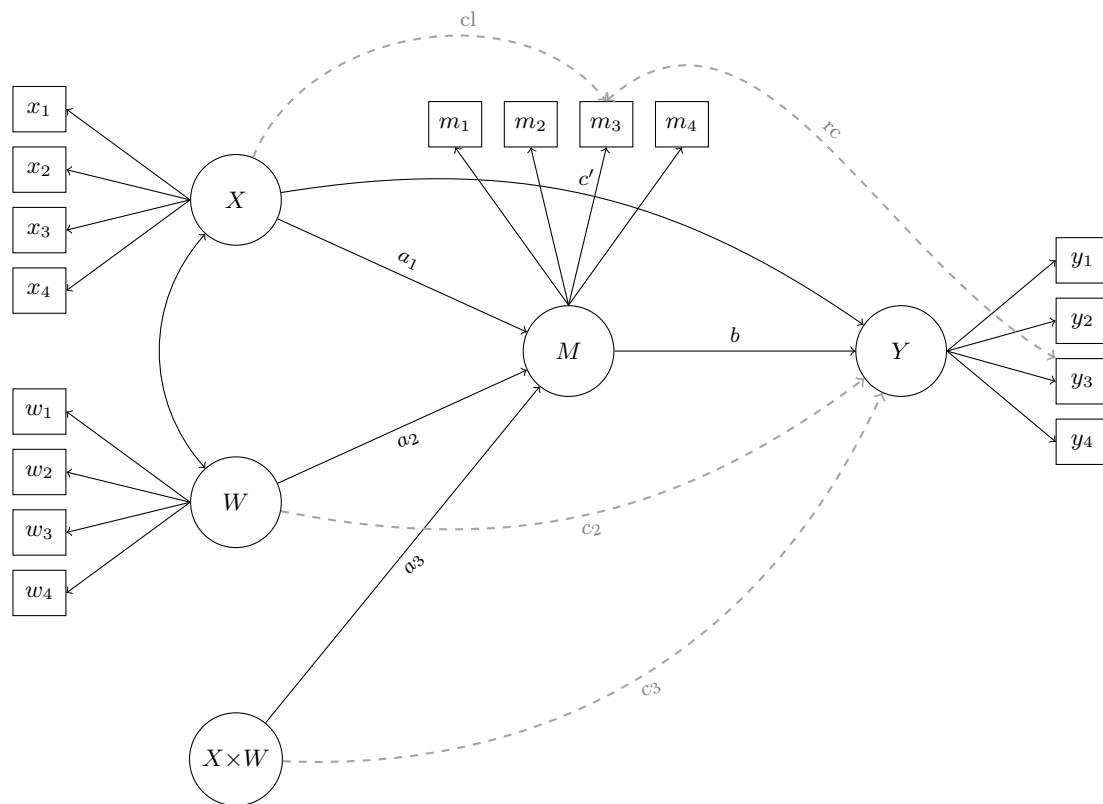
Table 2*Subset of the Performance Measures for Evaluating Parameter Estimates*

Performance measure	Empirical calculation	MCSE
<i>Relative criteria</i>		
Relative bias	$\frac{\bar{B}}{\theta} - 1$	$\sqrt{S_B^2 / (K\theta^2)}$
Relative RMSE	$\frac{1}{ \theta } \sqrt{\frac{1}{K} \sum_{k=1}^K (B_k - \theta)^2}$	$\frac{1}{ \theta } \sqrt{\frac{K-1}{K} \sum_{j=1}^K (\text{RMSE}_{(j)} - \text{RMSE})^2}$
<i>Standard errors</i>		
Mean SE	$\overline{SE} = \frac{1}{K} \sum_{k=1}^K \widehat{SE}_k$	—
SE/SD ratio	$\overline{SE} / S_B$	—
Relative bias of variance estimator	$\rho = \overline{SE^2} / S_B^2$	$\sqrt{\frac{K-1}{K} \sum_{j=1}^K (\rho_{(j)} - \rho)^2}$
<i>Coverage</i>		
Coverage	$c = \frac{1}{K} \sum_{k=1}^K I(L_k \leq \theta \leq U_k)$	$\sqrt{c(1-c)/K}$
<i>Hypothesis testing</i>		
Rejection rate	$r = \frac{1}{K} \sum_{k=1}^K I(0 \notin [L_k, U_k])$	$\sqrt{r(1-r)/K}$

Note. B_k is the estimate of parameter θ in replication k ($k = 1, \dots, K$ usable replications), $\bar{B}$ its mean, S_B^2 its sample variance, and S_B its standard deviation; $\text{RMSE} = \sqrt{\frac{1}{K} \sum_{k=1}^K (B_k - \theta)^2}$; $\widehat{SE}_k$ is the estimated standard error, $\overline{SE^2} = \frac{1}{K} \sum_{k=1}^K \widehat{SE}_k^2$, and L_k, U_k the confidence-interval limits; c and r denote coverage and rejection rate, and ρ the relative bias of the variance estimator. The subscript (j) marks a leave-one-out jackknife replicate. The MCSEs for relative RMSE and the relative bias of the variance estimator are computed by jackknife; the others are closed-form, and dashes mark MCSEs that are not computed. All measures were computed with the *simhelpers* package.

Figure 1

Population (Data-Generating) Model for the Simulation Study.



Note. *Note.* X , W , M , and Y are latent variables, each measured by four congeneric indicators; circles are latent variables and $X \times W$ is the latent interaction. Black elements form the correctly specified model that every estimator fits (a_1 , a_2 , a_3 , b , c' , and the X – W covariance). Grey elements are the misspecifications present in the population but omitted from the fitted model: two in the structural part, an omitted moderator effect on the outcome (c_2) and an omitted moderated direct effect (c_3); and two in the measurement part, an omitted cross-loading of X on m_3 (c_1) and an omitted residual covariance between m_3 and y_3 (rc). Residual variances and intercepts are not shown.

Figure 2
Relative Bias of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3

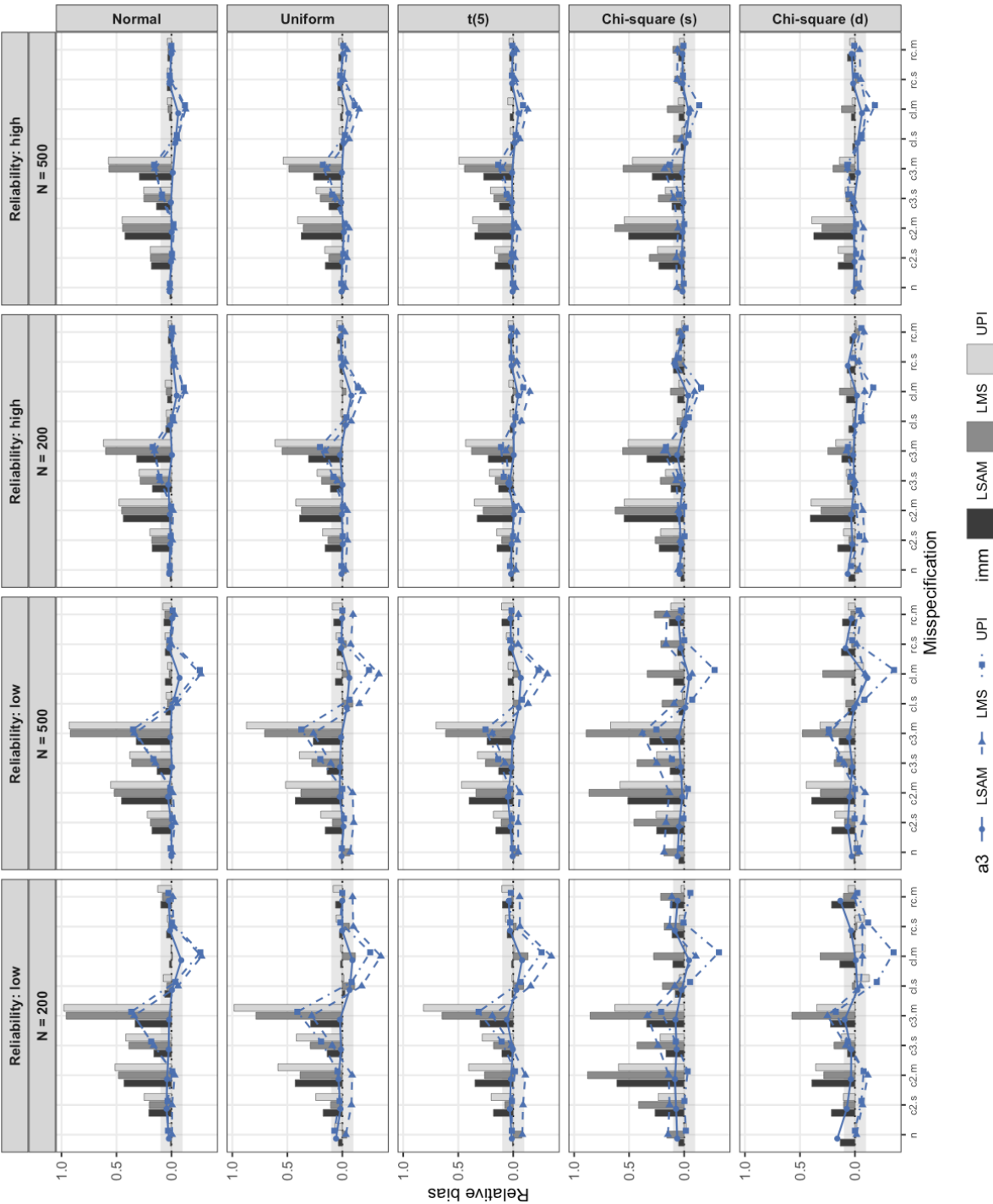


Figure 3

Relative RMSE of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3

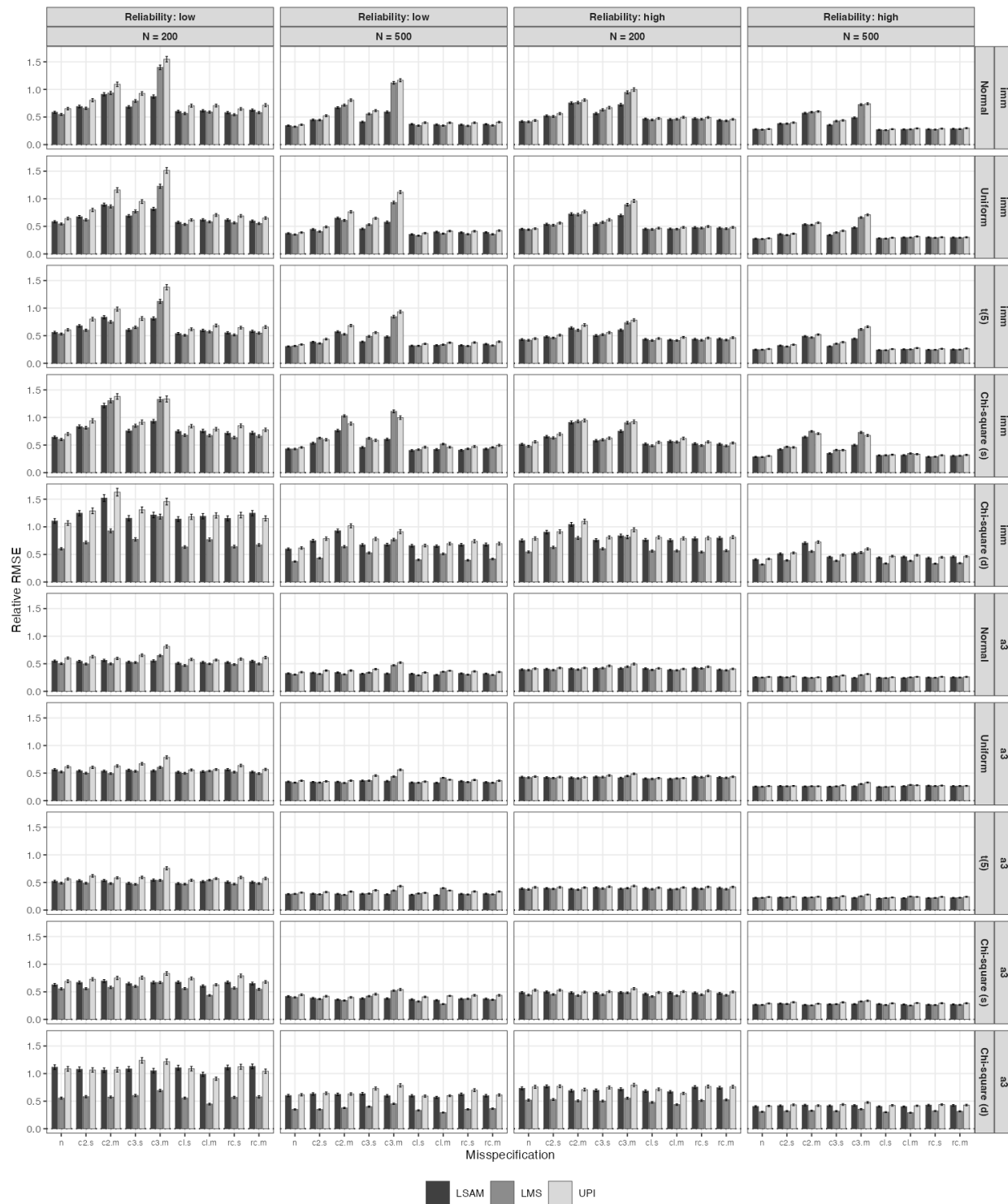


Figure 4
Coverage of the Index of Moderated Mediation

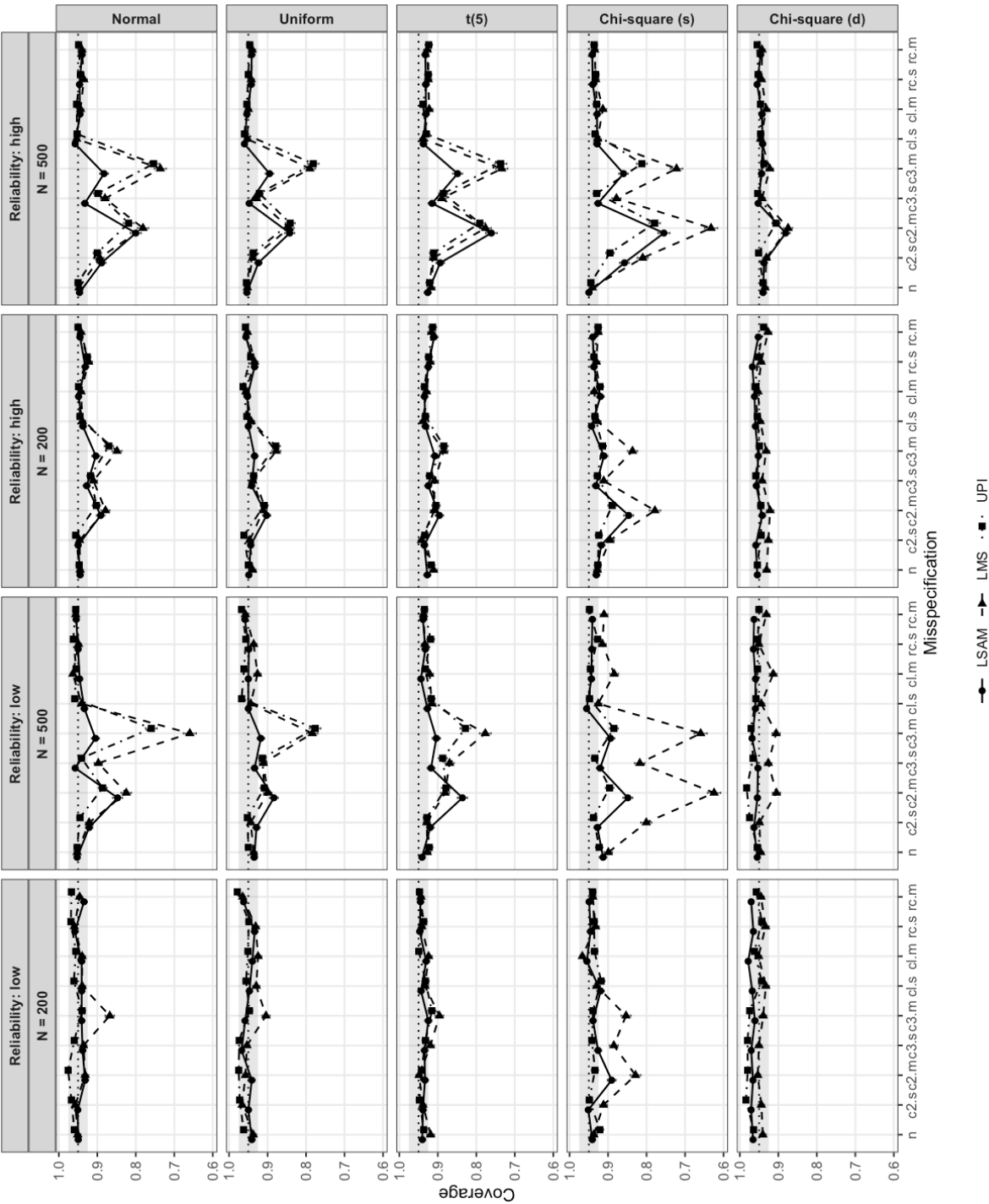


Figure 5
Coverage of the Interaction Coefficient a_3

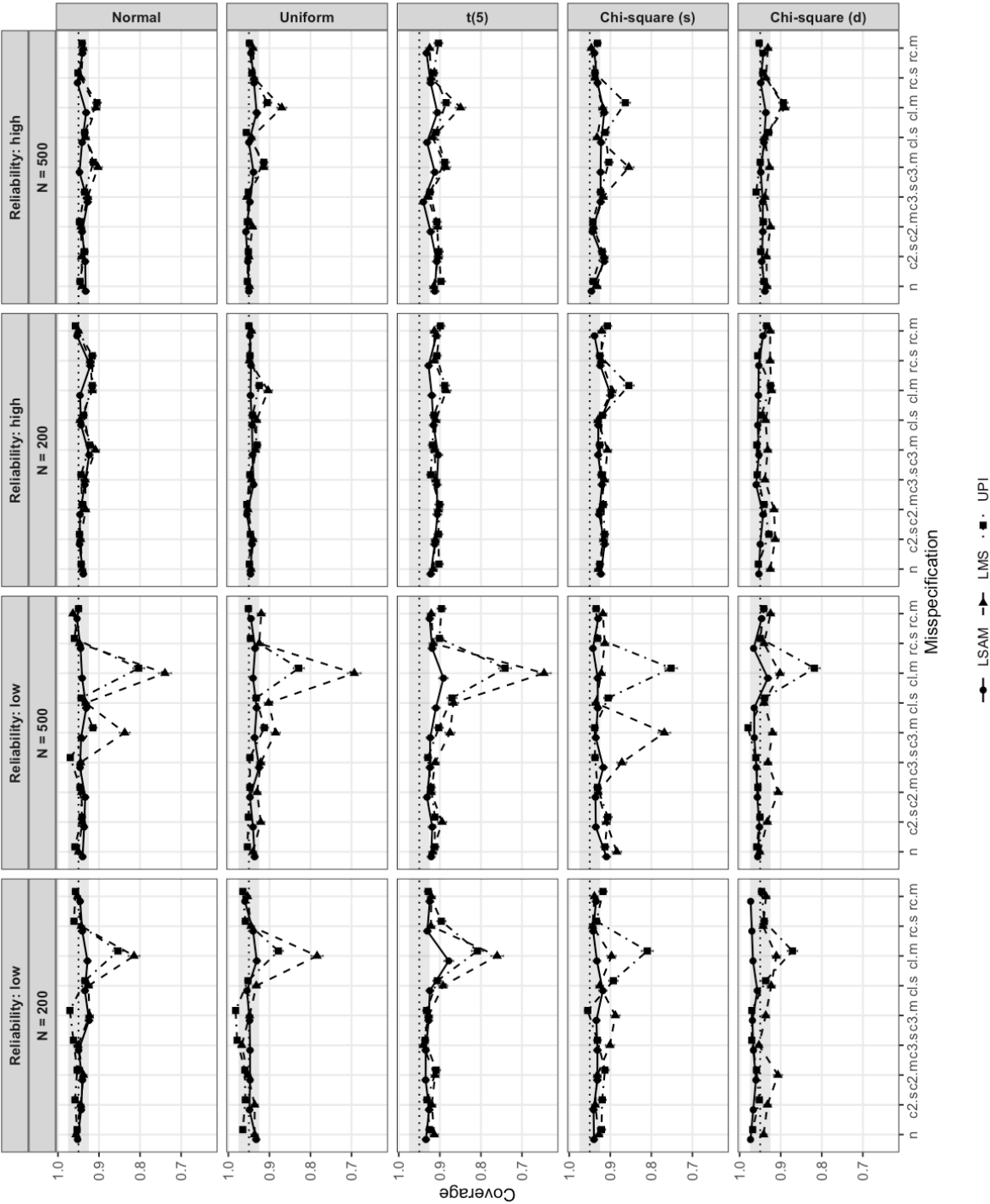


Figure 6
SE/SD of the Index of Moderated Mediation

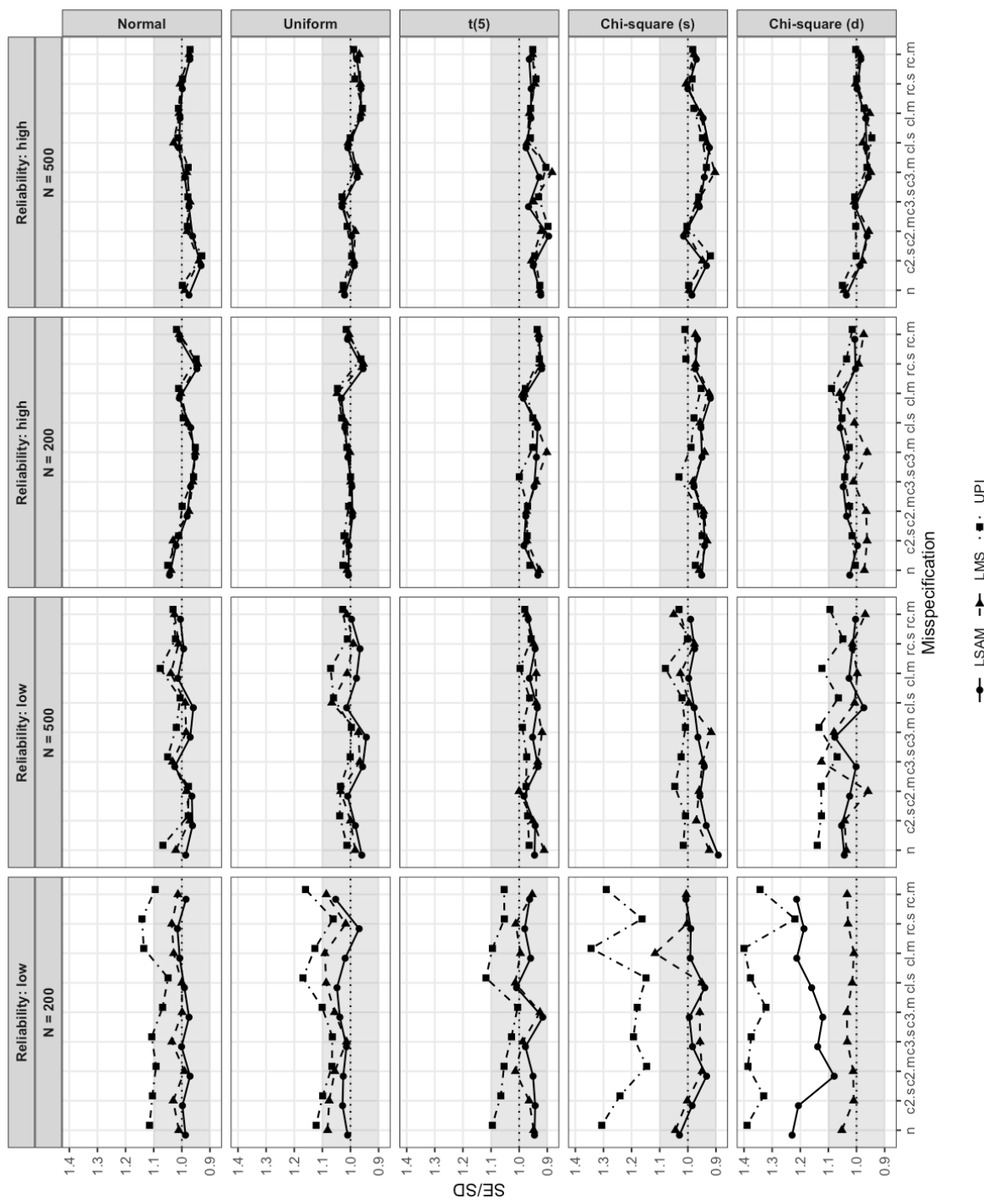


Figure 7
SE/SD of the Interaction Coefficient a_3

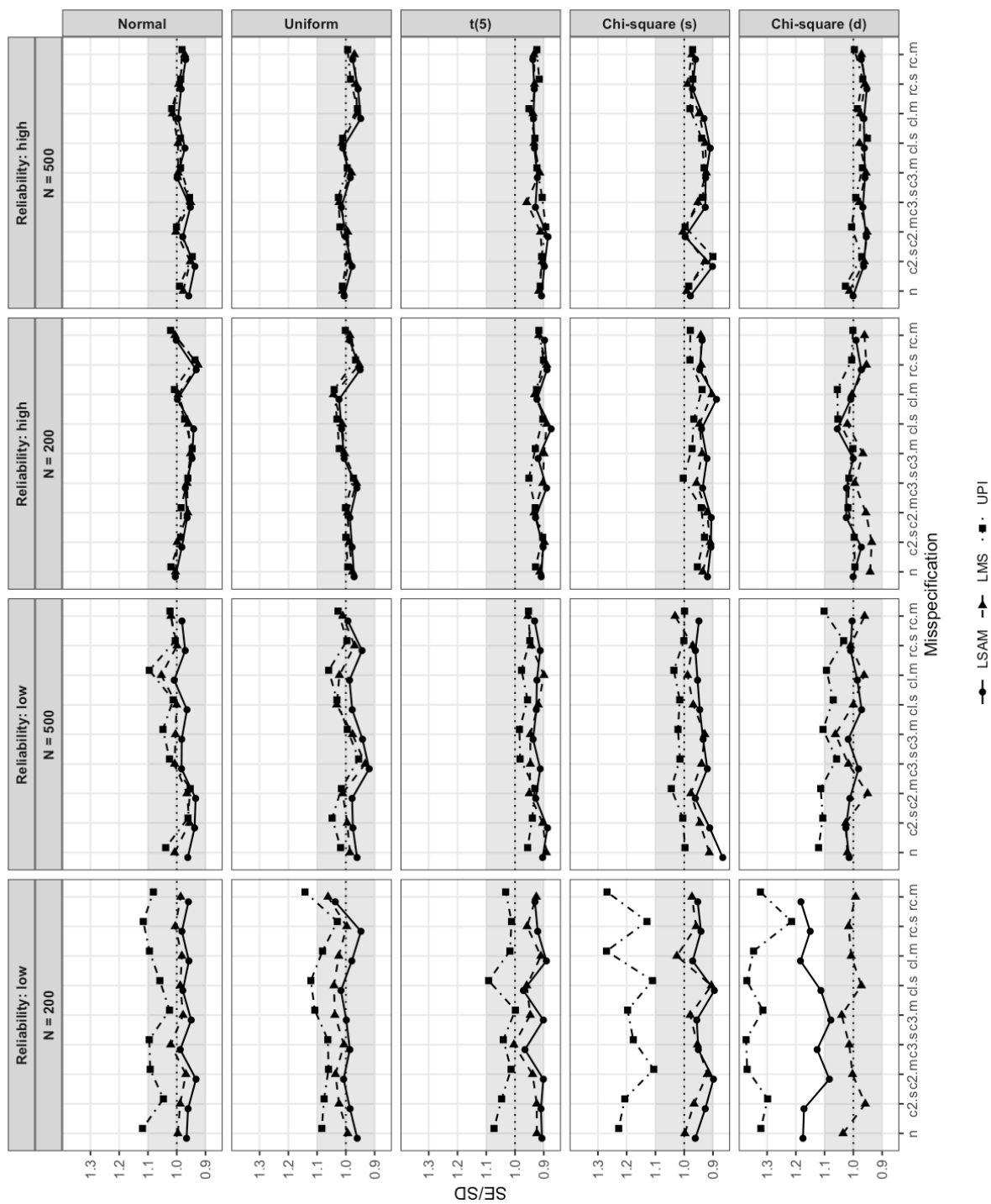


Figure 8
Relative Bias of the Variance Estimator of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3 (High Reliability)

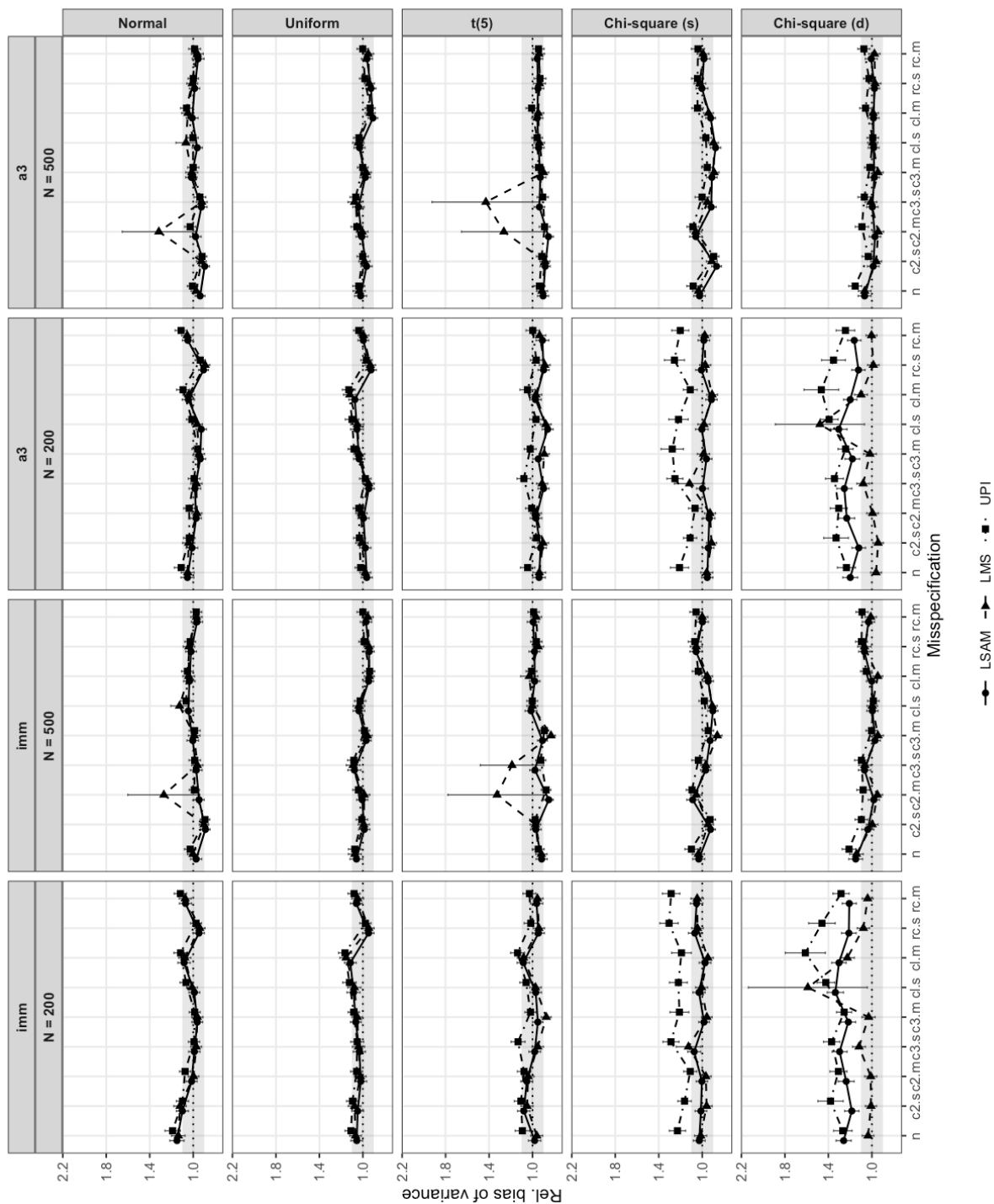


Figure 9

Type I Error of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3

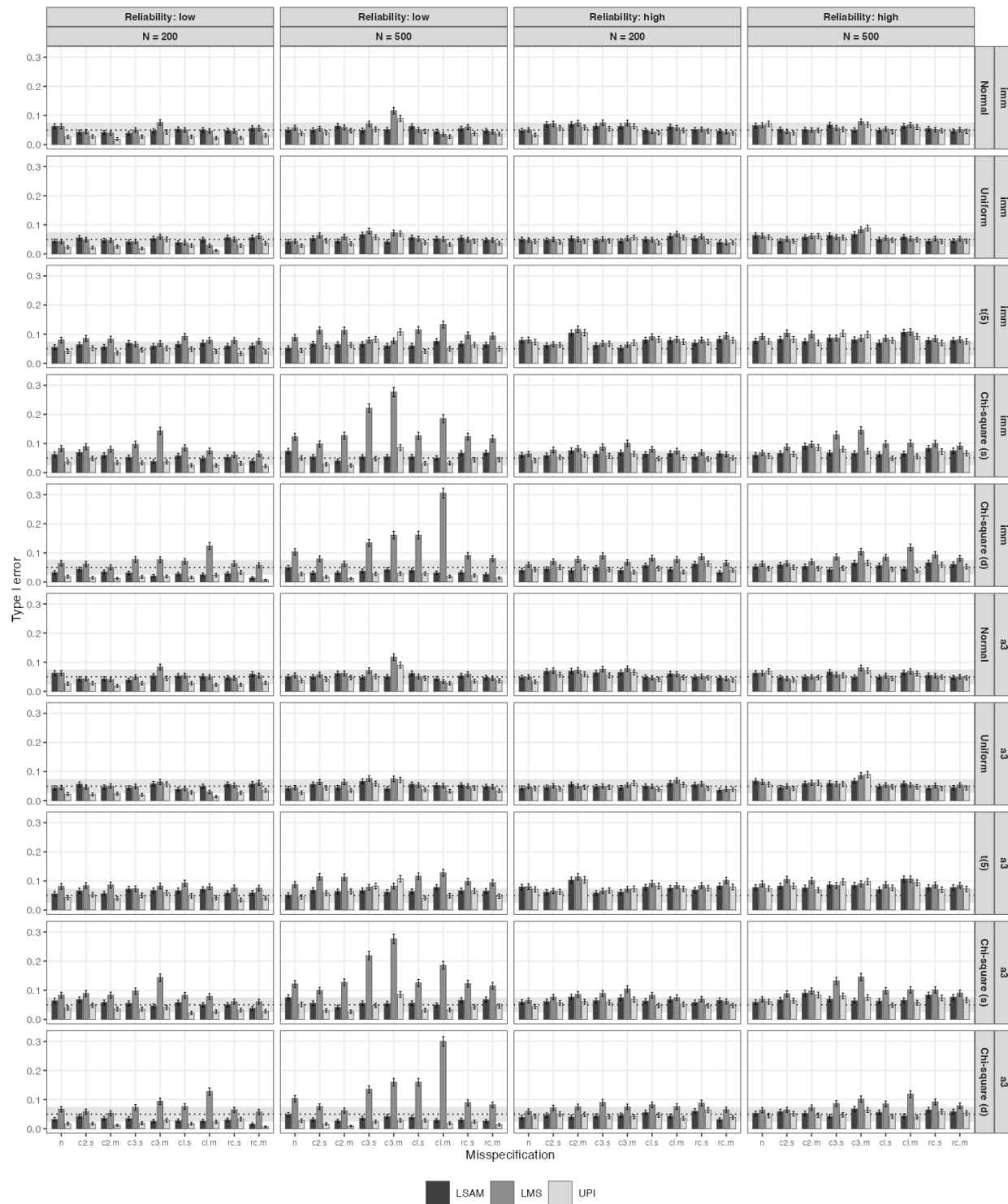


Figure 10

Power for the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3 at $a_3 = 0.2$

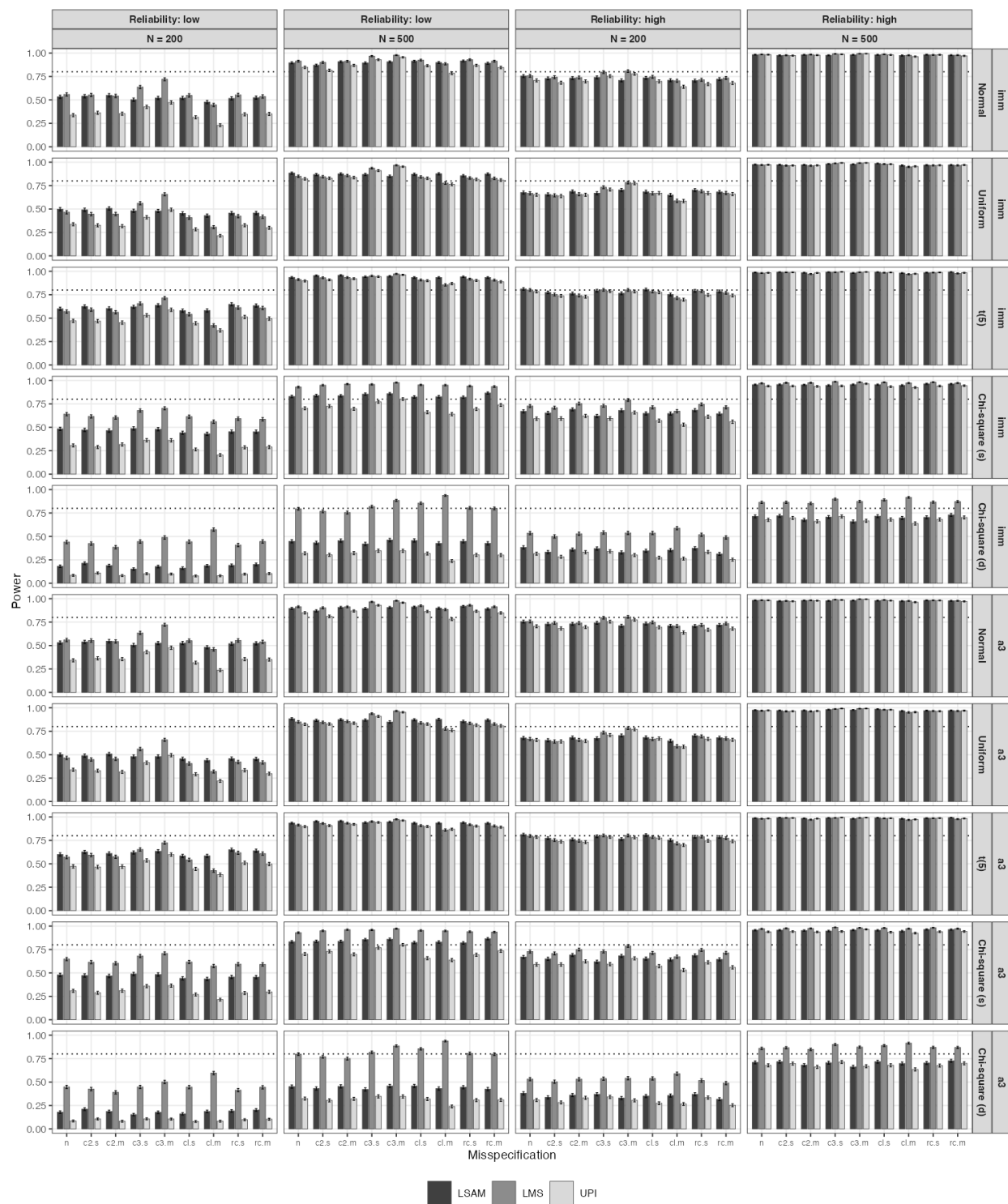


Figure 11
Relative Bias of the Structural Coefficients (a_1 , a_2 , b , c') at High Reliability and $N = 500$

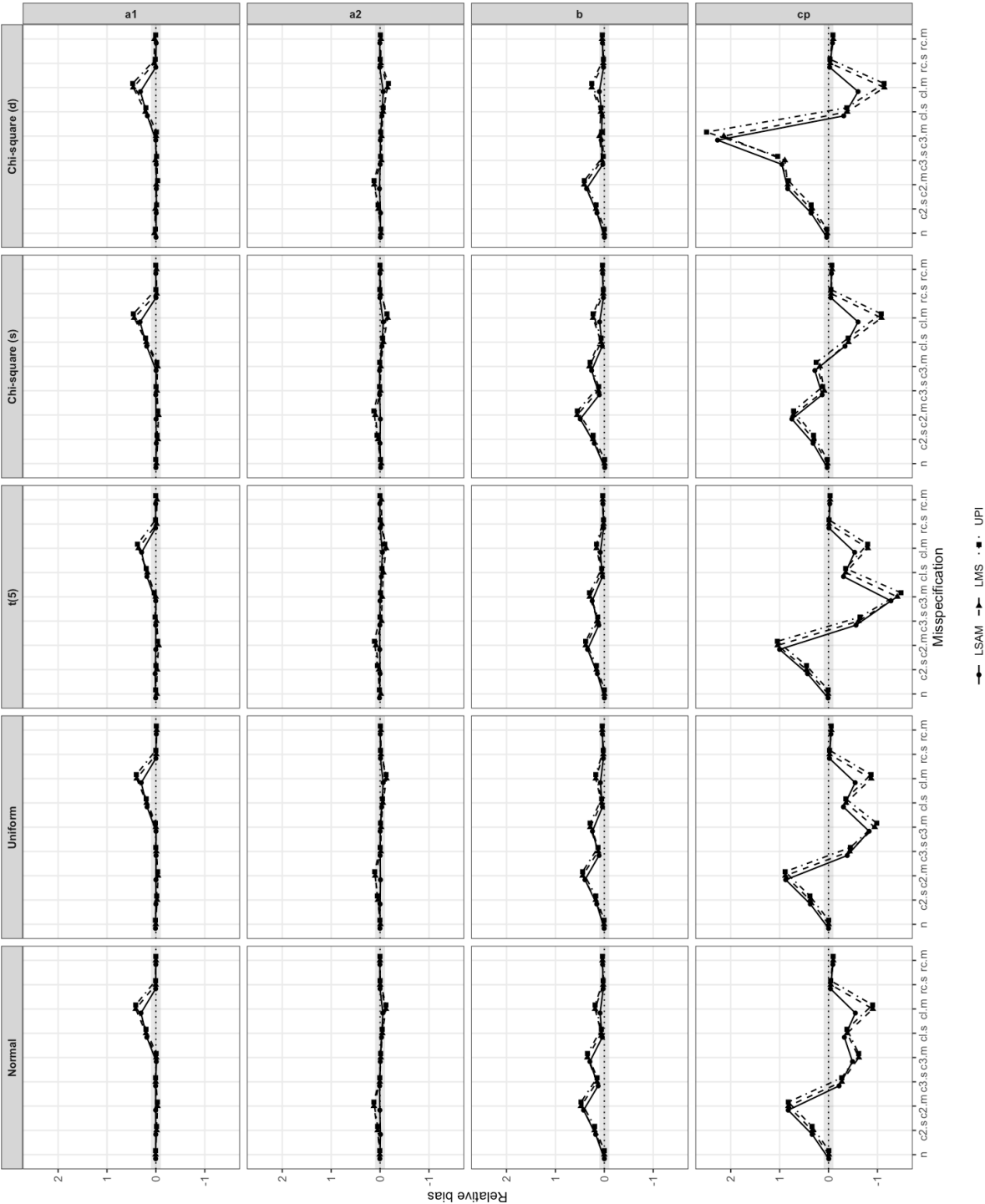


Figure 12
Coverage of the Structural Coefficients (a_1 , a_2 , b , c') at High Reliability and $N = 500$

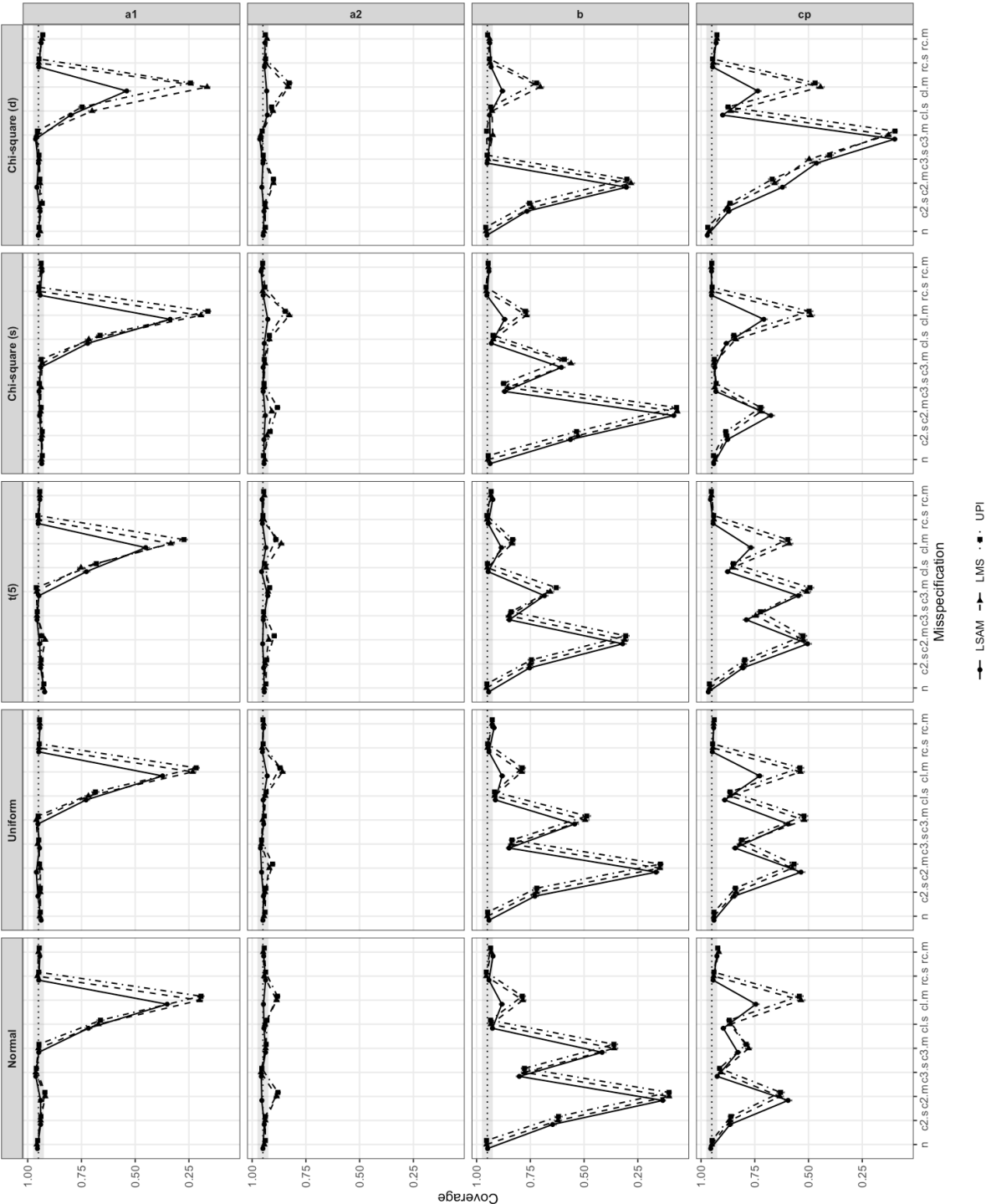


Figure 13
SE/SD of the Structural Coefficients (a_1 , a_2 , b , c') at High Reliability and $N = 500$

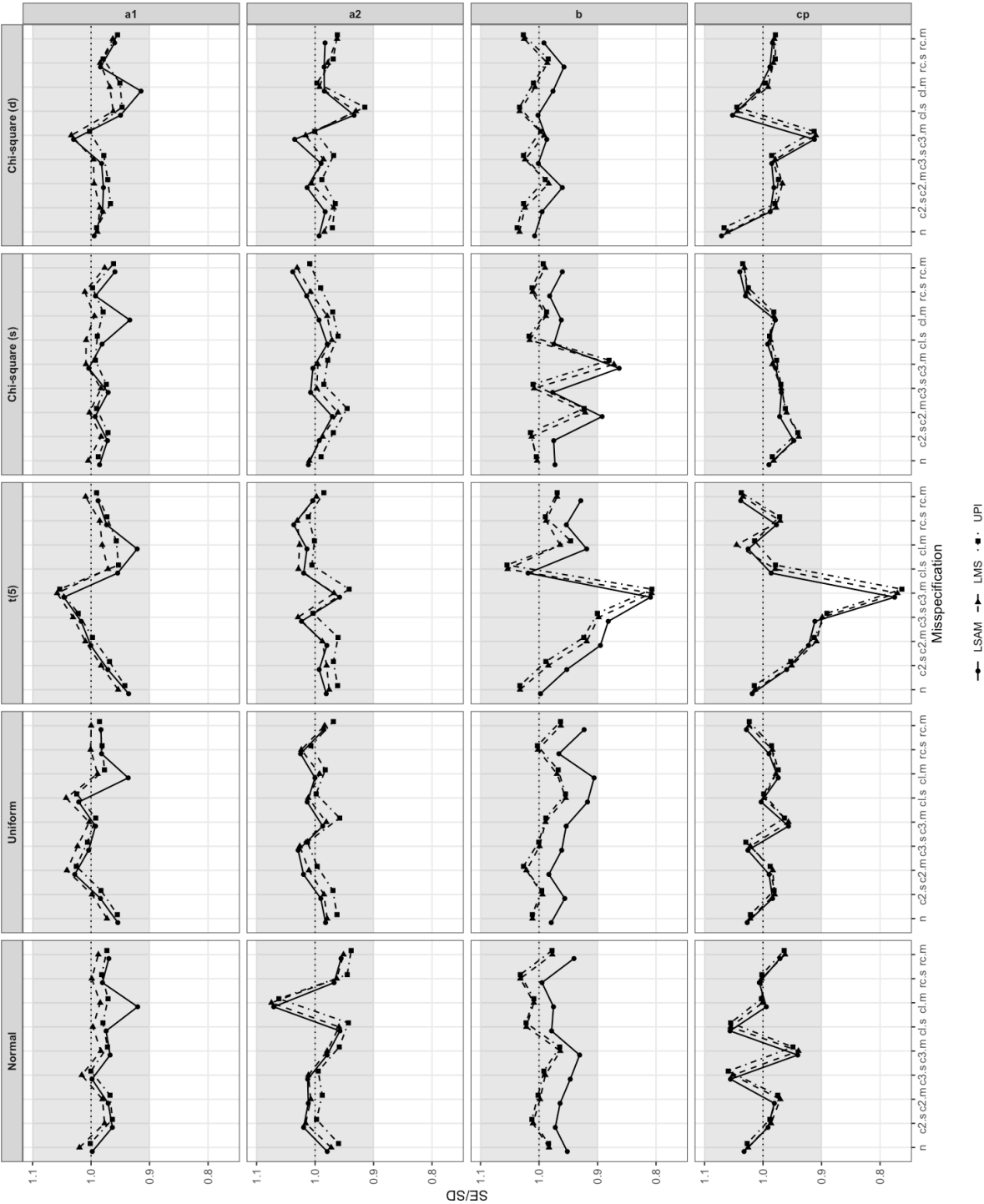
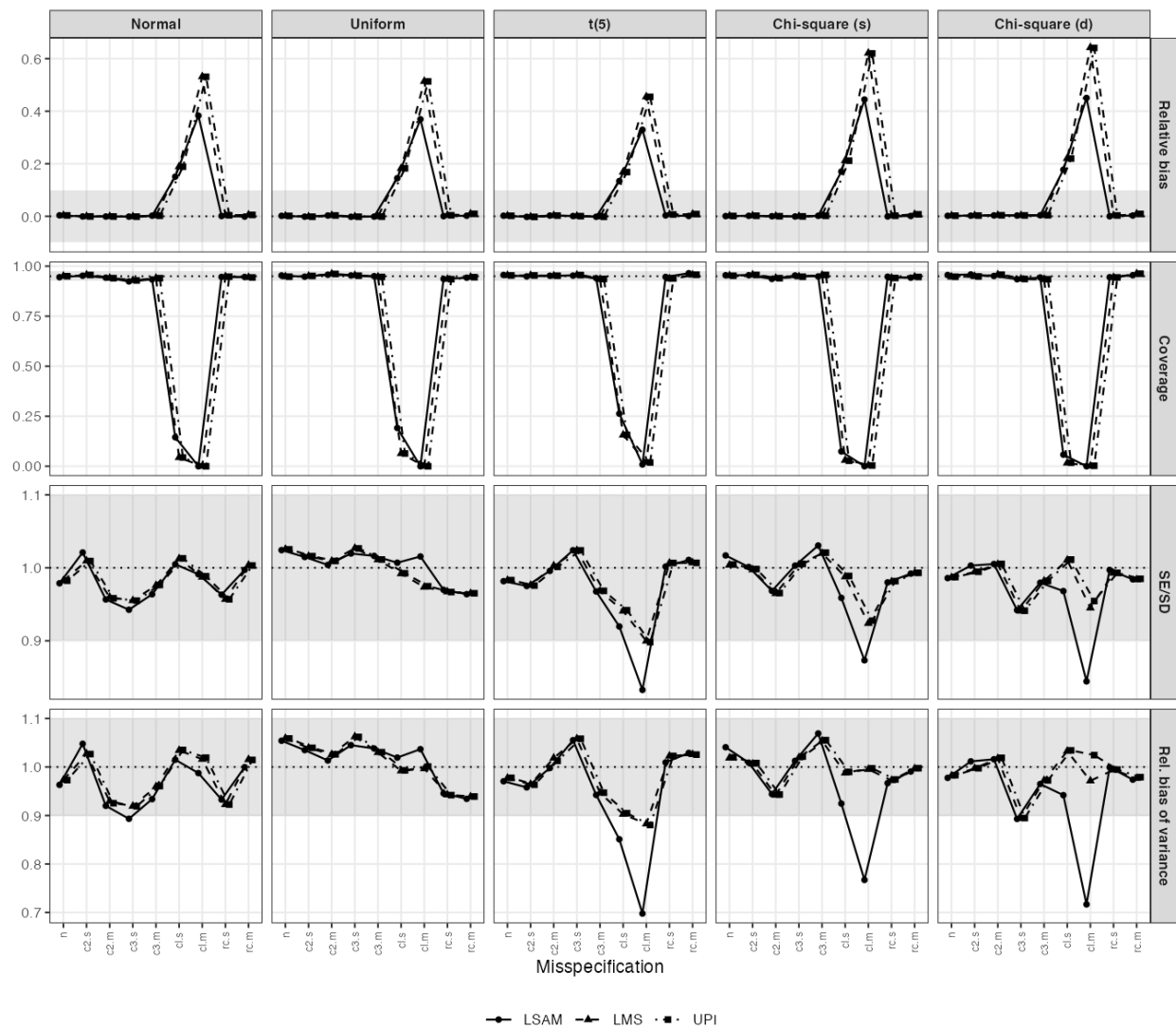


Figure 14

Performance of the Cross-Loaded Mediator Indicator Loading (m_3) Across All Measures at High Reliability and $N = 500$



Appendix A

Calibration and Data Generation

This appendix gives the formulas behind the data-generating mechanism. The full derivations, with worked numerical examples and even more details, are in the project repository.

Structural residual variances

The disturbance variances ψ_M, ψ_Y are fixed so that each structural equation explains a fraction $R^2 = 0.30$ of its outcome's variance at the central interaction level $a_3 = 0.2$ under the normal baseline. Writing the systematic part of M as $S_M = a_1X + a_2W + a_3(XW)$, its variance needs only second moments,

$$\text{Var}(S_M) = a_1^2 + a_2^2 + 2a_1a_2\rho + a_3^2(1 + \rho^2),$$

and the disturbance is the remainder that brings the signal share to R^2 ,

$$\psi_M = \text{Var}(S_M) \frac{1 - R^2}{R^2}, \quad \psi_Y = \text{Var}(bM + c'X) \frac{1 - R^2}{R^2},$$

with $\text{Var}(bM + c'X) = b^2 \text{Var}(M) + c'^2 + 2bc'(a_1 + a_2\rho)$. Fixed at these values, the population variances of M and Y are closed-form functions of the parameters,

$$\text{Var}(M) = \text{Var}(S_M) + \psi_M, \quad \text{Var}(Y) = b^2 \text{Var}(M) + c'^2 + 2bc'(a_1 + a_2\rho) + \psi_Y,$$

evaluated once under the normal baseline and held fixed across distributions, so that R^2 and reliability drift only through the design factors.

Indicator reliabilities

Each latent $\eta \in \{X, W, M, Y\}$ has four indicators $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$ with $\lambda = (1, 0.8, 0.7, 0.6)$. To hit a target reliability $\omega \in \{0.5, 0.7\}$, the residual variance is set from the latent variance $V = \text{Var}(\eta)$,

$$\theta_{\eta j} = \lambda_j^2 V \frac{1 - \omega}{\omega}.$$

For X, W the variance is exactly 1; for M, Y it is the closed form above, fixed at the normal baseline, so the realized reliability drifts mildly across distributions.

Exogenous distributions

In every cell (X, W) has mean 0, variance 1, and correlation $\rho = 0.3$; only the shape changes. The uniform branch uses $U(-\sqrt{3}, \sqrt{3})$; the t_5 branch samples with covariance $\frac{5}{3} \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$ and divides by $\sqrt{5/3}$; the χ^2_2 branches center by 2 and scale by $\sqrt{4}$. `chisq_same` leaves both margins right-skewed, whereas `chisq_diff` samples at $-\rho$ and negates W , flipping its skew while restoring $\text{Cor}(X, W) = +\rho$. Because $\text{Var}(XW)$ is a fourth-moment quantity, it is not constant across these conditions (≈ 1.09 under normality, larger under t_5 and χ^2).

Calibrating the misspecification magnitudes

The “small” and “medium” labels are standardized coefficients, $t \in \{0.25, 0.50\}$, not raw ones. For an omitted predictor p added to a baseline outcome y_0 , the magnitude is the fully standardized regression coefficient of p ,

$$t = \frac{\beta \text{sd}(p)}{\text{sd}(y_0 + \beta p)}.$$

Squaring and substituting $\text{Var}(y_0 + \beta p) = \text{Var}(y_0) + \beta^2 \text{Var}(p) + 2\beta \text{Cov}(p, y_0)$ yields a quadratic $A\beta^2 + B\beta + C = 0$ with

$$A = \text{Var}(p) (1 - t^2), \quad B = -2t^2 \text{Cov}(p, y_0), \quad C = -t^2 \text{Var}(y_0).$$

Since $A > 0$ and $C < 0$ the roots have opposite sign; the positive omitted path is the positive root,

$$\beta = \frac{t^2 \text{Cov}(p, y_0) + \sqrt{t^4 \text{Cov}(p, y_0)^2 + t^2 (1 - t^2) \text{Var}(p) \text{Var}(y_0)}}{\text{Var}(p) (1 - t^2)}.$$

The three moments are estimated from one large ($n = 10^6$) sample per distribution and a_3 level. The predictor p and baseline y_0 differ by misspecification,

$$c2 : \quad p = W, \quad y_0 = bM + c'X + \varepsilon_Y,$$

$$c3 : \quad p = XW, \quad y_0 = bM + c'X + \varepsilon_Y,$$

$$c1 : \quad p = X, \quad y_0 = m_3 = \lambda_3 M + \delta_{m_3}.$$

c2 and c3 are reliability-independent; c1 depends on ω because its baseline m_3 carries measurement noise. rc never enters the calibration: it is a residual correlation and equals its target in every cell.

Holding Y reliability fixed under c2 and c3

Adding an omitted structural path inflates $\text{Var}(Y)$ by $\kappa = \text{Var}(y_0 + \beta p) / \text{Var}(y_0)$, which would raise Y 's indicator reliability as a side effect. Scaling the Y -indicator noise by the same κ makes signal and noise move together, so κ cancels and the reliability returns to its no-misspecification value,

$$\omega_j = \frac{\kappa \lambda_j^2 V}{\kappa \lambda_j^2 V + \kappa \theta_j} = \frac{\lambda_j^2 V}{\lambda_j^2 V + \theta_j}.$$

The latent effect size is untouched; only the Y indicators are made correspondingly noisier. The measurement-level misspecifications c1 and rc act on the indicators directly and are not compensated, since that contamination is itself the misspecification.

Appendix B

Relative Bias of the Variance Estimator at Low Reliability

At low reliability the relative bias of the variance estimator becomes unstable under heavy-tailed predictors and small samples: a few replications with extreme standard errors inflate the measure, most severely for LMS. These low-reliability cells are collected here so that the main-text figure (Figure 8) can resolve the high-reliability conditions, where the measure stays near one. Cells above ten are truncated.

Figure B1
Relative Bias of the Variance Estimator of the Index of Moderated Mediation (imm) and the Interaction Coefficient a_3 (Low Reliability)

